

Lyra v0.1 Phase 3 — Dual-RX UI Candidates

Working-group handout — three approaches to consider

Phase 0 (scaffolding), Phase 1 (protocol nddc=4 dispatch), and Phase 2 (RX2 audio routing with hard-left/hard-right stereo split) are now committed and operator-verified on a HL2+. RX2 is a working second receiver with its own DDC, demod chain, and audio output landing in the right ear.

Phase 3 is the UI surface that exposes the second receiver to the operator. The engine is done; this question is purely about how operators interact with two receivers at once. Below are the three real candidates. Each comes from a real-world precedent so we’re not picking a novel design in a vacuum.

Option A — Hybrid (focused-RX panel)

*Single control panels; focus indicator shows which RX is active. Switch via **Ctrl+1** / **Ctrl+2** or by clicking the RX’s frequency display.*

RX1: 7.074.000 MHz		[FOCUSED – orange border]	RX2: 15.000.000 MHz								
MODE + FILTER		Mode: USB	RX BW: 2.7k	(operates on RX1 – focused)							
DSP + AUDIO		AGC	NR	NF	LMS	SQ	APF	Vol	Bal	AF	(operates on RX1 – focused)

Pros

- Saves screen real estate — fits laptops + 1080p easily
- One mental model: “what RX am I focused on?”
- Currently the consensus plan’s default bet (\$6.2)

Cons

- Can't see RX1's AGC state while looking at RX2
- Switching costs a click or hotkey every time
- Foreign to Thetis operators who expect full duplicated view

Real-world example: EESDR / SunSDR series, some Apache Labs ANAN UIs.

Option B — Tabbed

Single DSP+AUDIO panel with **RX1 / RX2 tabs** at the top. Click a tab to swap which RX’s controls are shown. Both VFO displays stay live; only the controls panel changes.

RX1: 7.074.000 MHz		RX2: 15.000.000 MHz	
MODE + FILTER	Mode: USB	RX BW: 2.7k	(controls follow active tab)
DSP + AUDIO	RX1 RX2	(click a tab to switch)	
AGC	NR	NF	LMS SQ APF Vol Bal AF Mute

Pros

- Same screen-real-estate savings as Hybrid
- Explicit visible “you are looking at RX1” via active tab
- Familiar UX pattern — browsers, IDEs, file explorers
- No focus-model concept needed; what you see is what you control

Cons

- Still can’t see both RX’s settings at once
- Tab-switching also costs a click
- Two freq displays + one controls panel can feel “disconnected”

Real-world examples: Web-based SDRs (KiwisDR consoles), some Android SDR apps. Less common on desktop ham apps.

Option C — Thetis-duplicate (both RX panels visible)

Two full DSP+AUDIO + MODE+FILTER stacks visible simultaneously, side-by-side. No focus model. Operator clicks directly on whichever control they want to adjust.

RX1: 7.074.000 MHz		RX2: 15.000.000 MHz	
USB 2.7k		AM 6.0k	
AGC NR NF LMS		AGC NR NF LMS	
SQ APF Mute		SQ APF Mute	
Vol Bal AF		Vol Bal AF	
[spectrum / waterfall]		[spectrum / waterfall]	

Pros

- Sees BOTH receivers’ state at a glance
- No focus model, no tab-switching, no hotkeys
- Standard convention from Thetis / PowerSDR
- Best for active dual-RX (split QSO, ragchew + DX hunt)

Cons

- Needs more screen real estate (1920×1080 minimum, ideally wider)
- Two of everything in the UI = more code to maintain
- Cramped on mobile / smaller laptops
- May feel busy to single-RX operators (RX2 off most of the time)

Real-world examples: Thetis, PowerSDR, EESDR's dual-RX layout, Yaesu FTdx101D / FTdx10 physical front panels follow the same pattern with their Sub-RX controls.

EESDR v3 — A reference for dual-RX + SPLIT UX

ExpertSDR3 (SunSDR3 / Apache Labs ANAN) is currently the most mature dual-RX implementation in the ham SDR world. Their conventions are worth borrowing wherever they fit. Quotes below are from the official ExpertSDR3 user manual.

Architecture

“2 software RXs + SubRX for each of them — 4 slices total.” EESDR’s RX1 and RX2 are *independent main receivers*, can tune to different bands — that maps to v0.4 ANAN-class hardware. Each main RX has its own **SubRX**, a second filter slice on the *same band* as its parent. That’s the closer analog to what we’re building for HL2: both DDCs share one ADC + one BPF passband.

Important takeaway for HL2 Lyra: what we call “RX2” is structurally closer to EESDR’s *SubRX* than to their RX2. v0.4 multi-radio work (ANAN-class) will introduce real independent RX1/RX2 on different bands.

SPLIT UX in EESDR

“There are two TX indicators, one near VFO A frequency and another near VFO B frequency. The active one is marked red and the inactive is marked gray. The TX indicator in red denotes the frequency of the associated VFO which you’re transmitting on.”

“Click-on the SPLIT button to initiate a split state. When in split state you will receive on the VFO A frequency, and transmit on the VFO B frequency...” Right-click SPLIT to set per-mode shift offset (saved per AM / LSB / USB / CW). In CW with Break.In + SPLIT, key-down auto-routes TX to VFO B.

SubRX UX in EESDR (HL2-relevant)

“Click-on middle mouse button is switching the point and the click control on the panorama between VFO A and VFO B receivers. The TUNE A or TUNE B near the mouse cursor indicates which of - VFO A or VFO B - is currently controlled.”

Within ONE main RX panel, EESDR has a single set of DSP controls (AGC, mode, filter, NR, etc.). Both VFO LEDs are visible. **Middle-click on the spectrum** swaps which VFO the cursor + wheel + main knob targets, with a tiny “TUNE A” / “TUNE B” hint near the cursor for unambiguous feedback. A **SUB button** toggles the SubRX on/off. Band change auto-disables SubRX.

Patterns Lyra Phase 3 should consider adopting

- **Two TX indicators**, one per VFO LED (red = active TX, gray = inactive). Click the gray one to swap.
- **Middle-click on the spectrum** swaps active VFO (faster than Ctrl+1/2 hotkey alone).
- **TUNE A / TUNE B tooltip** near the cursor on hover to disambiguate at-a-glance.
- **SUB / RX2-enable button** as a clear toggle (not a buried menu).
- **Right-click SPLIT** = per-mode shift parameter menu (Lyra extension — useful for DXers).
- **A→B / B→A / SWAP** buttons (already in consensus plan §6.3).

SPLIT workflow comparison

SPLIT is the heaviest dual-RX use case — the operator is actively driving BOTH receivers simultaneously. How each option handles it differentiates them more than anything else. Scenario: DX station calling on 14.205 USB “listening up 5”. Operator tunes RX1 to 14.205 (DX), RX2 to 14.210 (pile-up + own TX freq), engages SPLIT, calls.

Common SPLIT state across all options

RX1: 14.205.000 [TX gray]	RX2: 14.210.000 [■ TX RED]	[SPLIT ✓]
---------------------------	----------------------------	-----------

TX indicator on VFO B is RED (active TX), gray on VFO A. SPLIT button engaged. EESDR convention. This part is shared across all three options.

Bump RX2 (call freq) up 100 Hz	Click RX2 LED → wheel (1 click + wheel)	Click RX2 LED → wheel (1 click + wheel)	Click in RX2 panel → wheel (1 click + wheel)
Toggle RX1 NR (DX is noisy)	Ctrl+1 → click NR → Ctrl+2 (3 actions)	Click RX1 tab → click NR → click RX2 tab (3 clicks)	Click RX1's NR button (1 click)
Change RX2 filter (CW pile-up uses narrower BW)	Ctrl+2 → change filter (2 actions)	Click RX2 tab → change filter (2 clicks)	Click RX2's filter (1 click)
Watch both AGC meters live	Cannot — only focused RX's meter shown	Cannot — only active tab's meter shown	Yes — both meters visible simultaneously

Green column = lowest friction for that workflow step. Option C wins all four because both RX panels are simultaneously visible + clickable.

Click-cost on its own doesn't tell the whole story

The table above measures *mechanical click cost* for each DSP adjustment on the unfocused RX. By that metric alone, Option C is unambiguously cheapest. But click-cost isn't the same as **operator value** — how often do you *actually* need to change RX2's NR or filter during real SPLIT operation?

Real DXpedition SPLIT workflow on RX2 is almost entirely tuning. RX1 (the DX-listening side) gets set up once with appropriate DSP and stays put. RX2 is a *tuning tool* — you're moving its freq around to chase the DX's listening edge, find clean spots between callers, and time your call. You rarely need to change RX2's NR / filter / AGC during the pile-up. That shifts the calculus — see the next section.

Recommended approach — Option A + EESDR enhancements

Working-group lean (Brent + Timmy on Discord): **Option A (Hybrid) is the right base.** Combine it with EESDR's smart active-VFO patterns to address the friction concerns without giving up the clean Lyra identity. End result: not a Thetis clone, not pure EESDR, deliberately different.

Layout

RX1: 14.205.000 MHz [TX gray] ← focused (orange border) RX2: 14.210.000 MHz [▲ TX RED]

MODE + FILTER Mode: USB RX BW: 2.7k [SUB] [SPLIT ✓] [A→B] [B→A] [SWAP]

DSP + AUDIO AGC NR NF LMS SQ APF Vol Bal AF Mute (*operates on focused RX*)

What you see at a glance

- **Both VFO LEDs always visible** — never lose track of either frequency.
- **Two TX indicators**, one beside each LED. **RED** = will TX here. Gray = inactive. SPLIT toggles which is red. Click gray to manually swap.
- **Orange focus border** on whichever VFO the wheel / keyboard currently controls.
- **Single DSP+AUDIO panel** operates on the focused RX — clean, uncluttered, laptop-friendly. NR / NF / LMS / SQ / APF / AGC / Mode / Filter all follow focus.
- **SUB / SPLIT / A→B / B→A / SWAP buttons** in the MODE+FILTER strip, where they're available without context-switching.

Per-RX controls (refinement — visible only when SUB is on)

The single-RX layout has one Vol slider, one Mute button, and a Balance slider in the DSP+AUDIO row.

When SUB is enabled (or SPLIT is on), this expands:

- **Vol-A / Vol-B** — two independent volume sliders. RX1 and RX2 produce wildly different signal levels (DX whisper on one ear, hot SWL station on the other) and the operator needs to balance them ear-by-ear. This is the one control that genuinely benefits from per-RX independence in real dual-RX use.
- **Mute-A / Mute-B** — per-RX mute. Operator can silence RX2 (e.g., between pile-up calls) without tearing down the whole dual-RX setup.
- **Balance** stays single — it's an overall L/R lean on the combined output, not per-RX. (Per-RX pan defaults to hard-left/hard-right via WDSP SetRXAPanelPan and isn't operator-exposed in Phase 3.)
- **AF Gain** stays single — it's the pre-AGC system reference (rare to need divergent values).

Engine support: Phase 2 already wires per-channel volume through WDSP's ``SetRXAPanelPan`` + the dual demod path in ``do\_demod\_wdsp\_dual``. Adding the second Vol slider is two lines of state on Radio + a conditional UI element — no audio-engine change needed.

Tuning controls (EESDR-inspired)

- **Click a VFO LED** → focus moves there. Wheel / keyboard now tune that VFO.
- **Middle-click on the panadapter** → swaps active VFO + brief TUNE A / TUNE B tooltip near the cursor for unambiguous feedback. **Verified unbound on Lyra's panadapter today — safe to claim.**
- **Ctrl+1 / Ctrl+2** hotkeys for keyboard-only users.

- **Right-click on the SPLIT button** = per-mode shift-offset menu (e.g., set “up 5 kHz” default for USB; the offset is saved per mode — AM / LSB / USB / CW). This is on the button widget, NOT the panadapter — panadapter’s right-click is reserved for notch / dB-scale menus and is unchanged.

Walkthrough 1 — RX2 as a listening monitor

Scenario: You're ragchewing on 40m (USB voice). A buddy said they'd be on WWV 15 MHz checking time signals + checking propagation. You want to monitor both at once. No SPLIT — if you transmitted, it would go out on VFO A's freq.

State diagram

RX1: 7.150.000 MHz [▲ TX RED] ← focused		RX2: 15.000.000 MHz [TX gray]	
Mode: USB RX BW: 2.7k [SUB ✓] [SPLIT off] [A→B] [B→A] [SWAP]			

Workflow

1. **Click the SUB button.** RX2 panel + LED light up. RX2 begins at its persisted last freq.
2. **Click VFO B's LED** → focus shifts to RX2 (orange border moves). DSP panel now shows RX2's mode / filter.
3. **Type 15000000 into VFO B** or wheel to 15.000.000.
4. (Optional) **change RX2's mode to AM** — since RX2 is focused, the mode menu operates on it. RX1 stays in USB.
5. You hear **40m USB ragchew in LEFT ear, WWV AM in RIGHT ear** (Phase 2 stereo split).
6. **Ctrl+1** (or click VFO A's LED) to refocus RX1 for the rest of the QSO. The TX indicator stayed on VFO A throughout — if you keyed, you'd TX on 7.150 (the ragchew freq), not on WWV.

Key safety property: in non-SPLIT mode, the RED TX indicator stays glued to VFO A. You'd never accidentally transmit on WWV because the visible indicator tells you so.

Walkthrough 2 — RX2 as the SPLIT TX freq (DX pile-up)

Scenario: Rare DX on 14.205 MHz USB calling “listening up 5”. You want to call them. RX1 = DX freq (14.205), RX2 = your call freq + the pile-up you’re trying to bust (around 14.210). SPLIT engaged so TX goes out on VFO B (your call freq), not VFO A (the DX’s own freq — transmitting there would be rude).

State diagram

RX1: 14.205.000 MHz [TX gray] ← focused		RX2: 14.210.000 MHz [▲ TX RED]	
Mode: USB RX BW: 2.7k [SUB ✓] [SPLIT ✓] [A→B] [B→A] [SWAP]			

Workflow

1. **Tune RX1 to 14.205** — click-to-tune on panadapter, or LED edit. RX1 has USB + your usual DX DSP setup (light NR, narrow filter, fast AGC, whatever you prefer for weak-signal voice).
2. **Click SUB** → RX2 enables.
3. **Middle-click on the panadapter at ~14.210** → VFO B follows the click; tooltip flashes “TUNE B” near the cursor; focus moves to VFO B briefly.
4. **Click SPLIT** (or right-click for “up 5” preset). VFO B’s TX indicator turns RED, VFO A’s goes gray. You can now key safely knowing TX is going to 14.210, not 14.205.
5. **Listen to both receivers:** DX in left ear (14.205), pile-up in right ear (around 14.210). RX2’s DSP is at defaults (USB / 2.7k) which is fine — you don’t need to fiddle it during pile-up busting.
- 5a. **Balance the ears** — pile-up is louder than the DX. The two Vol sliders (Vol-A / Vol-B, visible because SUB is on) let you drop RX2 to a comfortable monitor level without losing the DX signal on RX1. Or Mute-B briefly between calls to focus the ear on RX1.
6. **Hunt for a clean call spot:** middle-click on the panadapter wherever the pile-up has a gap. VFO B updates. Repeat every few seconds as the pile-up shifts.
7. **Time your call** — listen for the DX to finish with the previous caller, then PTT. TX goes out on VFO B (14.210, your spot).
8. **You get through** — DX comes back with your call sign on 14.205 (RX1 left ear). You exchange reports, QSO done.
9. **Click SPLIT again to disengage** — TX returns to VFO A. **Or click SUB** to turn off RX2 entirely and reclaim the right-ear channel.

What if you DO need to change RX1’s NR mid-pile-up?

Rare in practice, but here’s the cost: RX1 is already focused (you tuned it first). The DSP panel is showing RX1’s settings. **Just click the NR button.** Single click. The extra cost only appears if you’ve middle-clicked over to RX2 recently (which sets focus to RX2). In that case it’s Ctrl+1 then click NR — 2 actions, not 3.

Compared to the relentless tuning-RX2 cadence of pile-up busting (where Option A is one click + wheel, same as Option C), this very-occasional DSP fiddle isn’t the dominant ergonomic cost. The earlier 4-row table over-emphasized it.

Honest observations for the working group

1. Screen size matters.

If most Lyra operators run 1920×1080 or wider, Thetis-duplicate is natural. If many run 1080p / 1440×900 laptops, Hybrid or Tabbed wins.

2. Workflow matters more.

Operators who actively work split (DX pile-ups, contesting) want to see both RX's at all times. Operators who use RX2 as a "background monitor" can live with the focus model.

3. Migration friction.

A Thetis / PowerSDR user moving to Lyra will *expect* the duplicate model. The first time they hit a focus model, it'll feel "wrong" until they retrain. The Hybrid bet says training cost is worth the screen savings; the Duplicate bet says no — give them what they already know.

4. Two-RX hardware reality.

HL2 / HL2+ are hard-locked to "RX1 and RX2 share one ADC, tuned within the same active band-filter passband." That's different from a 7000DLE or ANAN G2 where each RX is genuinely independent with wider front ends. For HL2 operators, a fully-duplicated panel may waste space showing RX2 settings that *typically* mirror RX1's. For ANAN / Brick operators with true per-RX flexibility, it's genuinely useful.

What we're asking the working group

The working-group lean today: **Option A (Hybrid) + EESDR active-VFO enhancements**, per the walkthroughs above. Phase 2's engine work supports A / B / C equally — the underlying second WDSP channel + audio combiner doesn't change. Phase 3 is purely the operator surface.

Specific input that would help most:

- **Does the Walkthrough 1 (RX2 listening monitor) flow feel right?** Specifically: does click-SUB / click-LED-to-focus / wheel-tune match how you'd expect to use a second RX?
- **Does the Walkthrough 2 (SPLIT pile-up) flow feel right?** Specifically: middle-click-to-tune-VFO-B + red-TX-indicator-on-active-VFO + SPLIT toggle. Anything missing that you'd hit in real DX pile-up operation?
- **Any per-RX setting you'd genuinely want independent** (and thus annoying with one-DSP-panel-per-focused-RX)? Per-RX Volume + Balance + Mute are already planned for Phase 3 even with Hybrid layout — what else?
- **Coming from Thetis / PowerSDR / EESDR — what specific habit will be hardest to give up?** Helps calibrate where to add EESDR-style discoverability shortcuts vs. where to deliberately stay Lyra-different.
- **Veto any of the EESDR-borrowed patterns?** (middle-click swap / TUNE A/B tooltip / red-gray TX indicators / right-click SPLIT shift menu). Anything feel wrong on second look?